

Skills Summary

Getting the Bounds, the Signs and the Variable Right

Use this reference while completing your worksheet or when studying.

Skill 1 — Bound order carries a sign

Reversal: $\int_a^b f = -\int_b^a f$. **Additivity:** $\int_a^c f + \int_c^b f = \int_a^b f$.

- The FTC subtracts in the order written: lower bound's value comes off the upper bound's value, even when the lower number is the larger one.
- If the variable sits at the *lower* bound of an accumulation function, flip it first — the flip is where the minus comes from.
- Under a substitution the bounds are values of the *old* variable. Convert them the moment you write $u = \dots$, or back-substitute to x before evaluating.

Example: Evaluate $\int_4^1 2x \, dx$.

Step 1. The bounds run backwards, so resist reordering them and subtract in the order written, $F(1) - F(4)$.

Step 2. $F(x) = x^2$, so the value is $1 - 16$
The integral is **-15**.

Try it: Evaluate $\int_5^2 3x^2 \, dx$.

check: -117

Skill 2 — Subtract the whole lower value

FTC Part 2: $\int_a^b f(x) \, dx = F(b) - F(a)$, where $F' = f$.

- Write $[F(x)]_a^b$ with real brackets, then $(F(b)) - (F(a))$ with real parentheses: the minus reaches every term of $F(a)$.
- Antiderivative signs to keep straight: $\int \sin x \, dx = -\cos x$ and $\int \cos x \, dx = \sin x$.
- Sign check: if the integrand is positive across the interval and the bounds increase, the answer must be positive.

Example: Evaluate $\int_1^2 (4x - 3) \, dx$.

Step 1. Antidifferentiate first and keep the lower value inside parentheses, since that is where the sign is usually lost.

Step 2. $[2x^2 - 3x]_1^2 = (8 - 6) - (2 - 3) = 2 - (-1)$
The integral is **3**.

Try it: Evaluate $\int_1^3 (2x - 5) dx$.

check: 97

Skill 3 — The variable of integration is a dummy

FTC Part 1: if $F(x) = \int_a^x f(t) dt$ then $F'(x) = f(x)$.

- Inside the integral, t is consumed: it cannot appear in the value of a definite integral, and a bound is never substituted into the integrand.
- F' is the integrand's *rule* with the bound variable in place of t — no antidifferentiating and no evaluating required.
- With the variable at the lower bound, flip and negate first (Skill 1).

Example: Let $F(x) = \int_1^x (t^2 + 3) dt$. Find $F'(4)$.

Step 1. The variable is already at the upper bound, so FTC Part 1 applies directly and no antiderivative is needed at all.

Step 2. $F'(x) = x^2 + 3$, so $F'(4) = 16 + 3$

The value is **19**.

Try it: Let $G(x) = \int_2^x (5t + 1) dt$. Find $G'(3)$.

check: 19

(!) Watch out: An answer that matches the expected size but not the sign is almost never an arithmetic error — check the bound order, the direction of an accumulation, and the sign of the antiderivative before re-doing the algebra.
